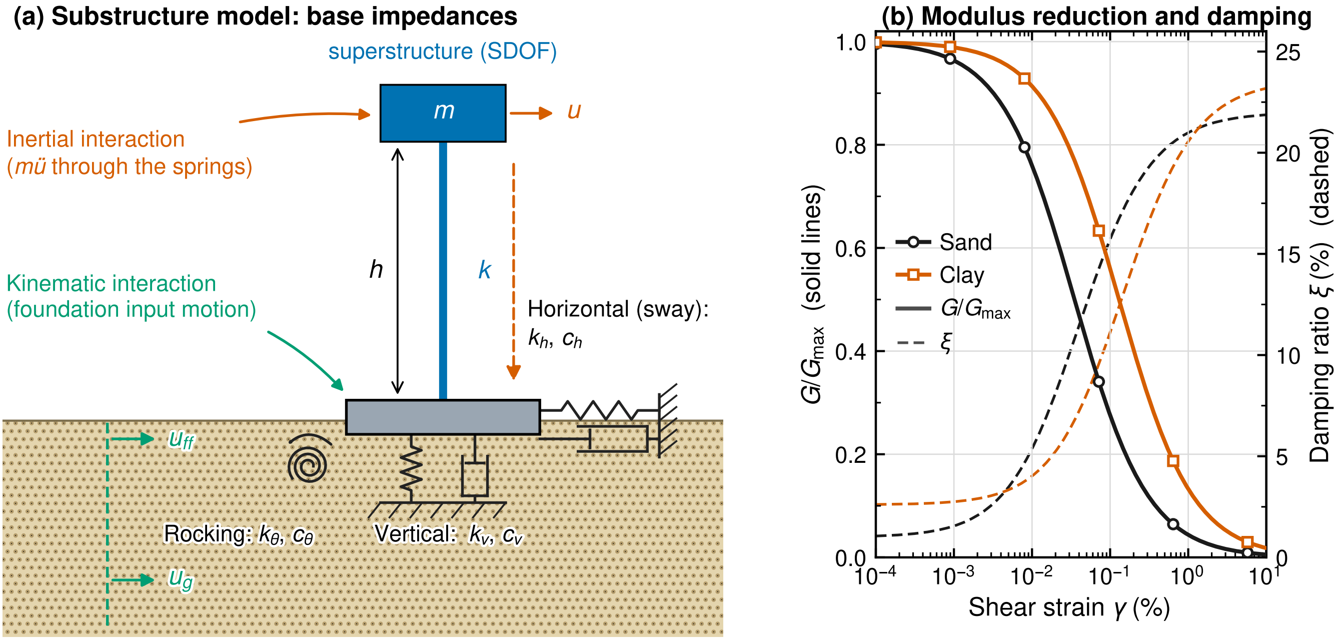
Soil–structure interaction and site response



Depiction: (a) substructure model of a rigid shallow footing on vertical, horizontal (sway) and rocking base impedances (spring–dashpot pairs), separating kinematic interaction (foundation input motion) from inertial interaction (*mü* transmitted through the springs); (b) modulus reduction G/G_{max} (solid, left axis) and damping ratio ξ (dashed, right axis) from a hyperbolic Hardin–Drnevich/Darendeli-type model, computed for a sand ($\gamma_r = 0.035$ %, $\xi_{min} = 1.0$ %) and a clay of PI $\approx$ 30 ($\gamma_r = 0.13$ %, $\xi_{min} = 2.6$ %) with $\xi_{max} = 21$ %; these curves set the strain-compatible soil-spring coefficients and the equivalent-linear site response. Software used for producing graph: Python 3.12.3 with Matplotlib 3.9.3 (pyplot, patches, lines, ticker, patheffects), NumPy 1.26.4 and SciPy 1.11.1. Source: authors.